Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Session	**2025-26 (EVEN)**	**Course Name**	**Operating Systems Lab**
Semester	**4**	**Course Code**	**23CT1402**
Roll No	**47**	**Name of Student**	**Dinesh More**

Practical Number	**03**
Course Outcome	1. Understand Computer System Configuration and Simulate system resources efficiently using Linux Commands (CO1) 2. Analyse operating system functionalities utilizing system calls, thread programming and process scheduling algorithms (CO2) 3. Apply Synchronization primitives to implement a Deadlock-free solution(CO3) 4. Simulate Disk scheduling, Memory allocation, File allocation, page replacement algorithms (CO4)
Aim	Aim: Execute Linux memory management system calls and Device management system calls.
Problem Definition	Execute Linux memory management system calls and Device management system calls.
Theory (100 words)	Theory:Linux memory management and device management system calls provide an interface between user programs and the operating system kernel. Memory management system calls are used to allocate, deallocate, and control memory during program execution. Calls such as brk() and sbrk() manage the process heap, while mmap() and munmap() are used to map files or devices into a process's address space for efficient memory access. These system calls help optimize memory usage and improve performance by enabling dynamic memory allocation and virtual memory management.

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	Device management system calls allow programs to interact with hardware devices through files. System calls like open(), read() write(), and close() are used to access devices such as disks, printers, and terminals. The ioctl() system call provides device-specific control operations. Together, memory and device management system calls ensure efficient resource utilization, controlled hardware access, and smooth communication between software applications and system hardware in a Linux operating system.
Procedure and Execution (100 Words)	Procedure Open the Linux terminal or Ubuntu using WSL on Windows. Create a C program file using the Vim editor. Write a C program that uses Linux memory management system calls such as mmap and munmap, and device management system calls such as open, read, write, and close. Save the program file and exit the editor. Compile the program using the GCC compiler. Execute the compiled program from the terminal. Observe how memory is allocated and released and how device files are accessed during execution.

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	Execution The program starts execution as a user process. Memory management system calls allocate memory dynamically during runtime. The allocated memory is used to store and process data. After usage, the memory is released back to the operating system. Device management system calls open a device or file and perform read and write operations. The close system call releases the device or file resource. The successful execution demonstrates efficient memory handling and proper device interaction in the Linux operating system.

	Code:
	#include <stdio.h>
	#include <stdlib.h>
	#include <fcntl.h>
	#include <unistd.h>
	
	int main(void) {
	  int fd;
	  char my_bucket[10];
	
	  fd = open("/dev/urandom", O_RDONLY);
	
	  if (fd < 0) {
	  perror("Failed to open");
	  return 1;
	  }
	
	  read(fd, my_bucket, 10);

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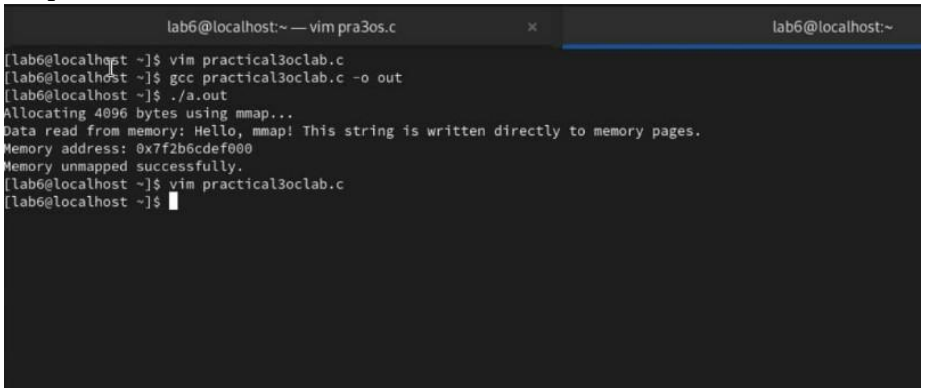
```
    printf("Random Hex Data: ");
    for (int i = 0; i < 10; i++) {
        printf("%02x ", (unsigned char)my_bucket[i]);
    }
    printf("\n");

    close(fd);

    return 0;
}
```

Output:



Output Analysis	**Data read from memory**
Link of student Github profile where lab assignment has been uploaded	https://github.com/More-Dinesh
Conclusion	Executed Linux management and device management system calls successfully
Plagiarism Report (Similarity index < 12%)	

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Date	4th February 2026